Infinity

ROTARY BURS
WITH CVDENTUS® DIAMOND



FR4-H



TR18-H



TR38-H

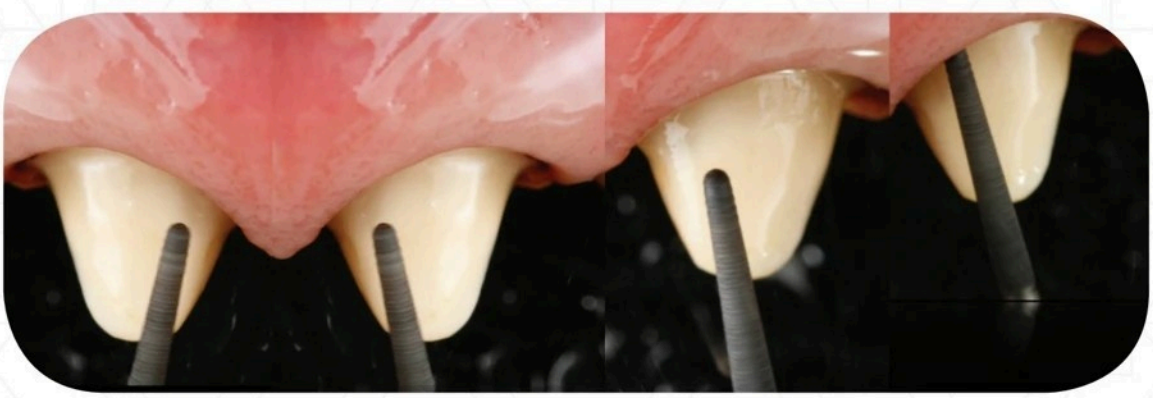
cvdentus

Benefits

- Durability 30x greater due to the extreme hardness of ~90 GPa (hardest material in nature)
- Thanks to CVD diamond, the homogeneous surface protects soft tissues, which are preserved when lightly touched
- Biologically compatible
- Vibration-free cutting
- Lower heat generation
- Higher cutting efficiency
- Excellent cost-benefit ratio
- Does not release metallic residues in the preparation
- Preparation and finishing in a single session, using the CVDentus® system together with Ultrasound









Product Identification

DIAMOND TIPS

Infinity tips are made with an innovative laboratory diamond production technology, originally developed for space-qualification equipment at INPE – the National Institute for Space Research. Infinity tips are genuinely national research products, an example of our country's capacity to generate technology.

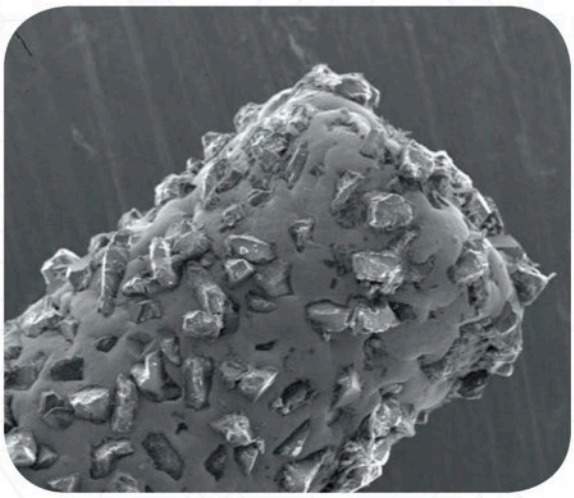
These tips, also known as burs, are dental instruments developed for dental procedures in general, performing grinding or cutting through rotation. To do so, the tip must be coupled to a handpiece attached to a rotary dental motor, and is compatible with several on the market. Infinity tips basically consist of a metal shank, one end of which is used for attachment to a handpiece for fixing to the motor; the other end is the active cutting part, covered with CVDENTUS® diamond, and cutting occurs through rotation of the tip around its own axis.

The active part of Infinity tips is fully coated with synthetic diamond grown by the CVD (Chemical Vapor Deposition) technique in gas-phase reactors, where the catalyst of the process is an electric current controlled by direct current (DC) sources. This causes heating and, consequently, the growth of this diamond on the molybdenum shank, and the roughness of the cutting surface can be controlled during this process, ranging from extremely fine to extremely coarse.

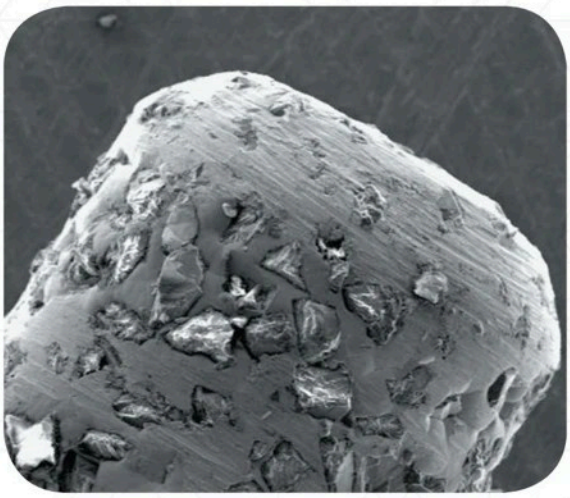
The fact that Infinity tips consist of a single diamond stone without edges reduces possible damage to soft tissue and does not allow contact with their metal parts, reducing the retention of dentinal residues, minimizing contamination risks and facilitating cleaning, with greater efficiency and greater durability. These characteristics and more are only possible because it is a tip with a completely closed diamond layer, without edges and highly adherent to the metal shank.

CVDentus Technology

CONVENTIONAL BUR FOR GRINDING

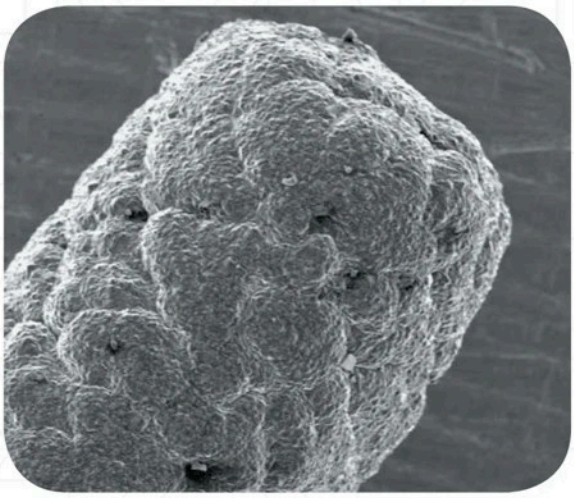


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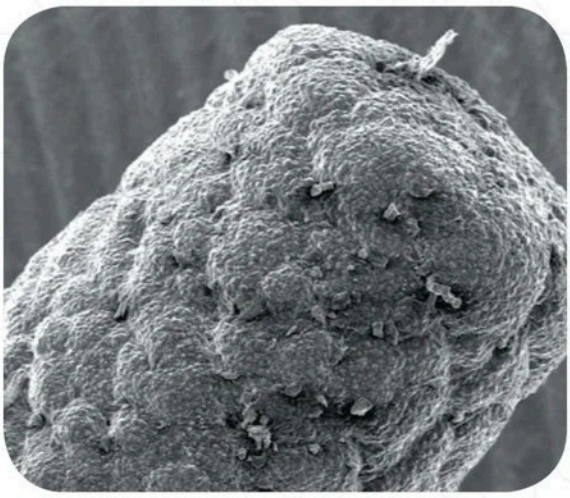


AFTER 5 PROCEDURES

CVDENTUS® BUR FOR GRINDING



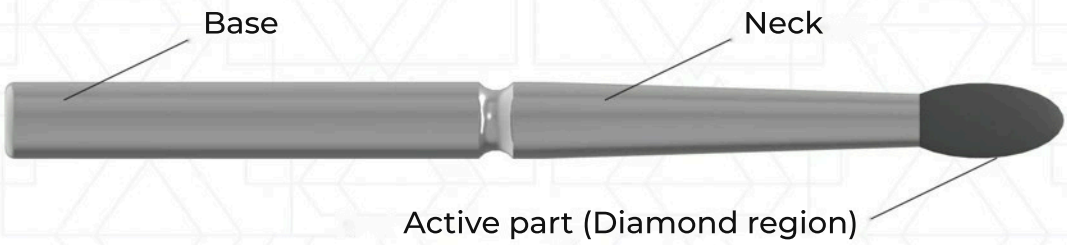
Unused



AFTER 100 PROCEDURES

Features

Figure 1 schematically shows the main parts of an Infinity tip.



Tip Composition

Pure diamond (active part – cutting region)
Pure molybdenum (neck and base)

Storage Conditions

As with other burs on the market, a dry, clean storage environment reserved exclusively for this purpose is recommended. Tips should be stored clean in their packaging, or autoclaved and stored in a specific container.

Cleaning and Sterilization

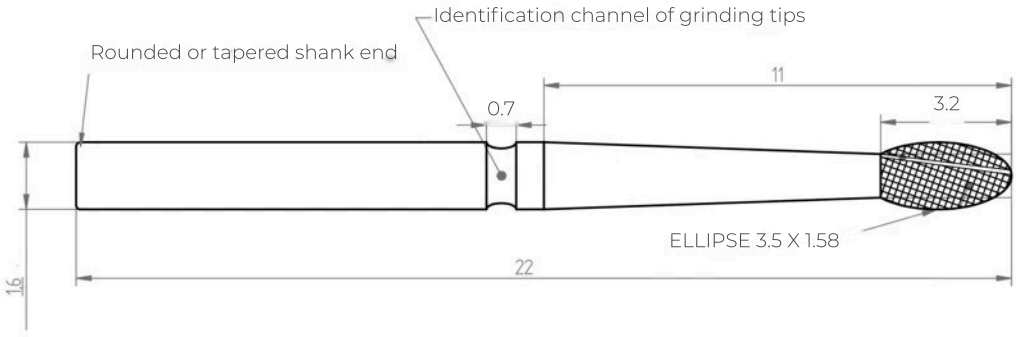
Infinity tips must be sterilized before each use by the autoclave process only.
1- Before starting the sterilization cycle, make sure the insert or bur is completely dry and free of residues.
2- Do NOT sterilize with glutaraldehyde or any other enzymatic material, to avoid corrosion of the metal shank.
3- Sterilize in an autoclave system using one of the following parameters:
121°C to 127°C (1 atm pressure) for 15 to 30 minutes;
132°C to 134°C (2 atm pressure) for 4 to 7 minutes.

Instructions for Use

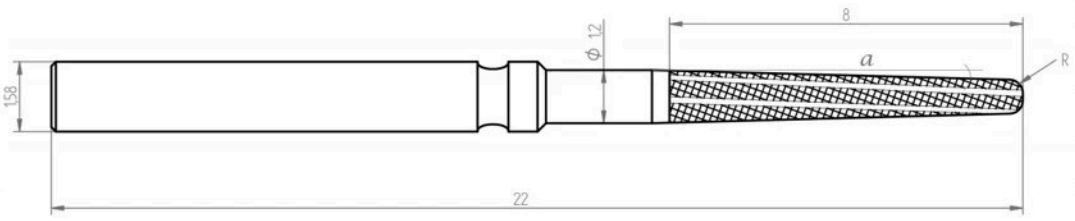
The assembly procedure must follow the instructions of the manufacturer of the handpieces or high-speed handpieces.

Indications

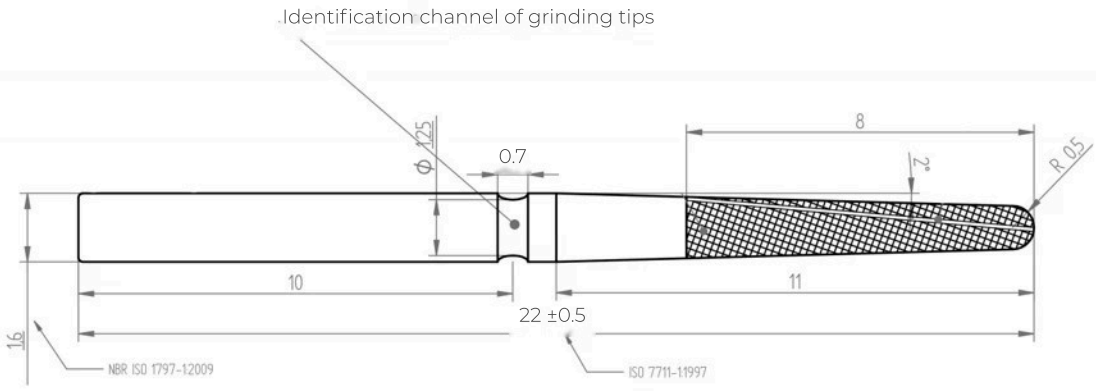
FR4-H - Ovoid shape (Restorative Dentistry: occlusal adjustment / Prosthodontics: prosthetic preparation, rounding of sharp angles for porcelain crowns, among others);



TR18-H - Ogival frustum shape (Prosthodontics: prosthetic repair, beveling of the finish line, among others);



TR38-H - Inverted cone shape (Restorative Dentistry: cavity preparation and removal of restorative material, among others).



Other information

- **Prosthodontics**

https://youtu.be/YvAx6ntmZRM

- **CVDentus dental burs cutting various materials**

https://youtu.be/CDlwEEnQ1IQ

- **Metal cutting with CVDentus Infinity burs**

https://youtu.be/rE7qa1ilwlk